Differential Equation of first Order

→ Differential Equation :-

An equation containing an independent variable, a dependent variable and the derivative of the dependent variable is called differential equation.

Eg; $\frac{dy}{dx} + 5y = e^x$

$\frac{dy}{dx}$ → derivative of dependent variable

y → Independent variable

e^x → dependent variable

→ Order of a differential equation :-

The order of the ~~he~~ highest-order derivative occuring in a differential equation is called the order of a differential equation.

Eg; (i) $\frac{d^2y}{dx^2} + y = 0$ ⇓ order → 2

(ii) $\frac{d^3y}{dx^3} + \left[\frac{d^2y}{dx^2}\right]^3 = e^x$ ⇓ order → 3

(iii) $\frac{d^2y}{dx^2} + \left[\frac{dy}{dx}\right]^4 = 0$ → order = 2

order ≠ power

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→ Degree of a differential equation :-

The power of the highest-order derivative occuring in a differential equation is called degree of differential equation.

Eg; ① $\frac{d^2y}{dx^2} + \left[\frac{dy}{dx}\right]^3 = 0$ order = 2, degree = 1

② $\left[\frac{d^2y}{dx^2}\right]^2 + \left[\frac{dy}{dx}\right]^4 = 0$ order = 2, degree = 2

③ $\frac{dy}{dx} + \sin\left[\frac{dy}{dx}\right] = 0$ order = 1, degree = not define

Note :- ① If any non-polynomial function, trigonometric function, exponential function, logarithmic function etc then, the degree of that function is not define.

② Order & Degree defined when differential equation are always natural number.

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Eg: ① $\sin\left(\frac{dy}{dx}\right) \longrightarrow$ Order = 1, degree not define

② $\cos\left(\frac{dy}{dx}\right) \longrightarrow$ order = 1, degree not define

③ $e^{(dy/dx)} \longrightarrow$ order = 1, degree not define

④ $\log\left(\frac{dy}{dx}\right) \longrightarrow$ order = 1, degree not define.

(*) Degree not define when the non-polynomial, trigonometric, etc functions are present in a equation, but if they are in multiple and sum of a derivative the degree define.

Eg; $\frac{dy}{dx} + \frac{d^2y}{dx^2} \sin y + y = 0$

Order = 2, degree = 1

Eg; $\frac{dy}{dx} + \log x + \sin x + \left[e^x \cdot \frac{d^3y}{dx^3}\right]^4 = 0$

order = 3
degree = 4

PYQ - objective

2023

Q. Degree of differential equation $x = \frac{dy}{dx} + \frac{dy}{dx} \cancel{\frac{d^2y}{dx^2}} + \left(\frac{dy}{dx}\right)^3$ is

(i) 1 (ii) 2 (iii) 3 (iv) 4

⇒ Correct option is (iii) 3.

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Q. The degree of the differential equation

$$y\frac{dx}{dy} - \left(\frac{dx}{dy}\right)^2 - \sin y\left(\frac{dx}{dy}\right)^3 - \cos x = 0$$ is

(i) 0 (ii) 1 (iii) 2 (iv) Cannot be determined

⇒ Degree = 3.

(option is not given)

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→ Method of solving first order & first degree differential equation :~

(i) Differential equation with variable seperable;

If the given differential equation in the form of

$$\frac{dy}{dx} = f(x) \cdot g(y)$$

then, seperate the both variable with same derivative; (x with x derivative & y with y derivative)

$$\Rightarrow \frac{dy}{dx} = f(x) \cdot g(y)$$

$$\Rightarrow \frac{1}{g(y)} dy = f(x)\, dx$$

Integrate both side, we have

$$\Rightarrow \int \frac{1}{g(y)} dy = \int f(x)\, dx$$

$$\Rightarrow G(y) = F(x) + C$$

Hence, this is the general soln

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where,

$G(y)$ = antiderivative of $g(y)$.

$F(x)$ = antiderivative of $f(x)$.

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Q.1. Solve

$$\frac{dy}{dx} = \frac{x+1}{2-y}$$

Sol:

Given,

$$\frac{dy}{dx} = \frac{x+1}{2-y}$$

$$\Rightarrow (2-y)\,dy = (x+1)\,dx$$

Integrate both side

$$\Rightarrow \int (2-y)\,dy = \int (x+1)\,dx$$

$$\Rightarrow 2y - \frac{y^2}{2} = \frac{x^2}{2} + x + C$$

$$\Rightarrow \frac{4y - y^2}{2} = \frac{x^2 + 2x + 2C}{2}$$

$$\Rightarrow 4y - y^2 = x^2 + 2x + C$$

Hence, solⁿ of differential equation

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~~Q.2. Solve $(x^2 - y^2)\,dx$~~

Q.2. Solve $\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y\cos y}$

$\Rightarrow$ Given,

$$\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y\cos y}$$

$$\Rightarrow (\sin y + y\cos y)\,dy = (2x\log x + x)\,dx$$

Integrate both side, we have

$$\Rightarrow \int(\sin y + y\cos y)\,dy = \int((2x\log x) + x)\,dx$$

$$\left[\text{Apply } \int I \cdot II\,dx = I\int II\,dx - \int\left[\left(\frac{d}{dx} I\right)\int II\,dx\right]dx\right]$$

$$\Rightarrow -\cos y + y\int\cos y\,dy - \int\left[\frac{d}{dy}y\int\cos y\,dy\right]dy =$$

$$2\log x\int x\,dx - \int\left[\frac{d}{dx}\log x\int x\,dx\right]dx + \frac{x^2}{2}$$

$+c$

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$$\Rightarrow -\cos y + y\sin y - \int \sin y\, dy = x^2 \log x - \int \frac{x}{2}\, dx + \frac{x^2}{2} + C$$

$$\Rightarrow -\cos y + y\sin y + \cos y = x^2 \log x - \frac{x^2}{4} + \frac{x^2}{2} + C$$

Q.3. Solve $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$

$\Rightarrow$ Given,

$$\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{+3x}}{e^{2y}} + \frac{x^2}{e^{2y}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{3x} + x^2}{e^{2y}}$$

$$\Rightarrow e^{2y}\, dy = (e^{3x} + x^2)\, dx$$

Integrate both side, we have

$$\Rightarrow \int e^{2y}\, dy = \int (e^{3x} + x^2)\, dx$$

$$\Rightarrow \frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + C$$

$$\Rightarrow \frac{e^{2y}}{2} = \frac{e^{3x} + x^3 + 3C}{3}$$

$$\Rightarrow 3e^{2y} = 2(e^{3x} + x^3 + 3C)$$

Q.4. $y\sqrt{(1-x^2)}\,dy + x\sqrt{(1-y^2)}\,dx = 0$

$\Rightarrow$ Given,

$$y\sqrt{1-x^2}\,dy + x\sqrt{1-y^2}\,dx = 0$$

$$\Rightarrow y\sqrt{1-x^2}\,dy = -x\sqrt{1-y^2}\,dx$$

$$\Rightarrow \frac{y}{\sqrt{1-y^2}}\,dy = -\frac{x}{\sqrt{1-x^2}}\,dx$$

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